

HW02 – Domain Testing on EShop

Student information

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I. Feature: Pool A - FR-03: Forgot password and password reset (two steps)

1.Requirement Analysys

Functional requirements summary

FR-03 defines a two-step password reset flow:

- Step 1: The user enters an already registered email address.
- The system generates a 6-digit OTP and presents it in the demo environment.
- Step 2: The user enters the OTP, a new password, and a confirmation of the new password.
- The new password must follow the same strength rules as FR-01.
- The two password fields must match.
- The OTP must be valid for the same email that initiated the reset.

User inputs

The main user inputs for this feature are:

- Email address (Step 1)
- OTP (Step 2)
- New password (Step 2)
- Confirm new password (Step 2)

Assumptions

- The new password uses the same strength rules defined in FR-01: minimum 8 characters, at least one uppercase, one lowercase, one digit, and one special character.
- OTP is a numeric 6-digit value, represented as a string in the API.
- OTP validity is tied to the specific email that requested the reset.
- The spec does not explicitly define an expiry window or one-time-use rule, but SEC-07 suggests these should be treated as expected constraints.

2. Input Domain Identification

Input 1: Email address

- Data type: String
- Valid domain: A syntactically valid email address that belongs to an existing registered user
- Invalid domain: Empty, malformed, unregistered, non-string values
- Constraints:
 - Must be present
 - Must follow email format
 - Must correspond to a registered account
- Dependencies:
 - Existing user database
 - Reset request initiated for that same email
- Assumptions:
 - The UI uses email input semantics and the backend validates format

Input 2: OTP

- Data type: String
- Valid domain: A 6-digit numeric code that matches the OTP generated for the current reset request
- Invalid domain: Wrong value, wrong length, non-numeric, expired/used, empty
- Constraints:
 - Must be exactly 6 digits
 - Must be associated with the current email reset request
- Dependencies:
 - The email that initiated reset
 - OTP generation record/session state
- Assumptions:
 - OTP is single-use and time-bounded, as implied by security requirements

Input 3: New password

- Data type: String
- Valid domain: A password that meets the strength rules and is not empty
- Invalid domain: Too short, missing required character classes, empty/whitespace, non-string
- Constraints:
 - Minimum 8 characters
 - At least one uppercase letter
 - At least one lowercase letter
 - At least one digit
 - At least one special character from the allowed set
- Dependencies:
 - Password confirmation field
 - Reset request context
- Assumptions:
 - Same rules from FR-01 apply

Input 4: Confirm new password

- Data type: String

- Valid domain: A value identical to the new password
- Invalid domain: Empty, different from the new password, non-string
- Constraints:
 - Must match the new password exactly
- Dependencies:
 - New password input
- Assumptions:
 - Confirmation is only a matching check, not an independent strength check

3. Equivalence Class Partitioning

Email address

- EC1: Valid registered email, e.g. user@domain.com
- EC2: Valid format but unregistered email
- EC3: Empty or whitespace-only input
- EC4: Malformed email format, such as missing @, missing domain, invalid characters
- EC5: Wrong data type, such as null or numeric input

Why these classes exist:

- They separate successful flows from different failure categories.
- They distinguish syntax problems from account-state problems.

OTP

- EC1: Correct 6-digit OTP for the current reset request
- EC2: Incorrect 6-digit OTP for the same email
- EC3: OTP with fewer than 6 digits
- EC4: OTP with more than 6 digits
- EC5: Non-numeric OTP
- EC6: Empty OTP
- EC7: OTP that is expired or already consumed

Why these classes exist:

- They cover normal valid input and all meaningful invalid input patterns related to length, format, and validity state.

New password

- EC1: Strong password meeting all rules
- EC2: Password shorter than 8 characters
- EC3: Password missing uppercase letter
- EC4: Password missing lowercase letter
- EC5: Password missing digit
- EC6: Password missing special character
- EC7: Empty or whitespace-only password

- EC8: Non-string input

Why these classes exist:

- Each class isolates a single rule violation, which is important for domain-driven validation.

Confirm new password

- EC1: Matches the new password exactly
- EC2: Empty or whitespace-only confirmation
- EC3: Different from the new password
- EC4: Non-string input

4. Test cases for domain testing

Test Case ID	Requirement	Input(s)	Test Data	Equivalence Class ID	Expected Result	Test Type (Valid / Invalid)
TC-FP-001	Step 1: Submit a registered email to request OTP	Email address	user@domain.com	EC1	System accepts the email, generates a 6-digit OTP, and moves to Step 2	Valid
TC-FP-002	Step 1: Submit a syntactically valid but unregistered email	Email address	unknownuser@domain.com	EC2	System rejects the request with an appropriate error message and does not continue the reset flow	Invalid
TC-FP-003	Step 1: Submit an empty email field	Email address	""	EC3	System rejects the request and prompts for an email	Invalid
TC-FP-004	Step 1: Submit whitespace-only email	Email address	" "	EC3	System rejects the input as invalid	Invalid
TC-FP-005	Step 1: Submit malformed	Email address	invalid-email	EC4	System rejects the input with a	Invalid

	email format				format validation error	
TC-FP-006	Step 1: Submit null email value	Email address	null	EC5	System rejects the request and shows a validation error	Invalid
TC-FP-007	Step 1: Submit non-string email input	Email address	12345	EC5	System rejects the input and does not proceed	Invalid
TC-FP-008	Step 1: Submit a valid email with special characters in the local part or domain	Email address	user+tag@sub.domain.com	EC1	System accepts the email if it is syntactically valid and linked to a registered account	Valid
TC-FP-009	Step 2: Submit valid OTP and matching password fields	OTP, New password, Confirm new password	OTP: 123456; New password: StrongPass 1!; Confirm: StrongPass 1!	OTP EC1, Password EC1, Confirm EC1	Password reset succeeds and the user is notified of success	Valid
TC-FP-010	Step 2: Submit an incorrect OTP	OTP	999999	OTP EC2	System rejects the reset and shows an OTP validation error	Invalid
TC-FP-011	Step 2: Submit OTP shorter than 6 digits	OTP	12345	OTP EC3	System rejects the OTP as too short	Invalid
TC-FP-012	Step 2: Submit OTP longer than 6 digits	OTP	1234567	OTP EC4	System rejects the OTP as too long	Invalid
TC-FP-013	Step 2: Submit non-numeric OTP	OTP	ABC123	OTP EC5	System rejects the OTP as invalid format	Invalid

TC-FP-014	Step 2: Submit empty OTP	OTP	""	OTP EC6	System rejects the submission and requires OTP entry	Invalid
TC-FP-015	Step 2: Submit an expired or already used OTP	OTP	Previously expired/consumed OTP	OTP EC7	System rejects the reset and states that the OTP is no longer valid	Invalid
TC-FP-016	Step 2: Submit OTP that belongs to a different email context	OTP + email context	OTP generated for another email address	Dependency validation	System rejects the reset because the OTP is not valid for the current email	Invalid
TC-FP-017	Step 2: Submit a new password shorter than 8 characters	New password	Ab1!	Password EC2	System rejects the password as too short	Invalid
TC-FP-018	Step 2: Submit password missing uppercase letter	New password	strongpass1!	Password EC3	System rejects the password because it lacks an uppercase character	Invalid
TC-FP-019	Step 2: Submit password missing lowercase letter	New password	STRONGPASS1!	Password EC4	System rejects the password because it lacks a lowercase character	Invalid
TC-FP-020	Step 2: Submit password missing digit	New password	StrongPass!	Password EC5	System rejects the password because it lacks a numeric character	Invalid
TC-FP-021	Step 2: Submit password	New password	StrongPass1	Password EC6	System rejects the password	Invalid

	missing special character				because it lacks a special character	
TC-FP-022	Step 2: Submit empty new password	New password	""	Password EC7	System rejects the submission and requires a password	Invalid
TC-FP-023	Step 2: Submit whitespace-only new password	New password	" "	Password EC7	System rejects the password as invalid	Invalid
TC-FP-024	Step 2: Submit null new password	New password	null	Password EC8	System rejects the password and shows a validation error	Invalid
TC-FP-025	Step 2: Submit empty confirm password	Confirm new password	""	Confirm EC2	System rejects the submission and asks the user to confirm the password	Invalid
TC-FP-026	Step 2: Submit confirm password that does not match the new password	Confirm new password	New password: StrongPass 1!; Confirm: StrongPass 2!	Confirm EC3	System rejects the reset and reports that the password values do not match	Invalid
TC-FP-027	Step 2: Submit non-string confirm password	Confirm new password	123456	Confirm EC4	System rejects the input and prevents reset	Invalid
TC-FP-028	Step 2: Submit multiple invalid inputs together	OTP, New password, Confirm new password	OTP: 000000; New password: weak; Confirm: different	OTP EC2, Password EC2/EC3/E C6, Confirm EC3	System reports all relevant validation errors and does not reset the password	Invalid
TC-FP-029	Step 2: Submit form	OTP, New password,	OTP omitted;	OTP EC6	System blocks	Invalid

	with a missing required field	Confirm new password	New password provided; Confirm provided		submission and highlights the missing required OTP field	
TC-FP-030	Step 2: Submit password with special characters and matching confirmation	New password, Confirm new password	New: P@ssw0rd!; Confirm: P@ssw0rd!	Password EC1, Confirm EC1	System accepts the password and allows password reset	Valid

5. Boundary Value Analysis

Inputs with measurable boundaries

The measurable boundaries identified from the requirement are:

- OTP length: exactly 6 digits
- New password length: minimum 8 characters

The email address and confirm password do not have measurable numeric or ordered boundaries in the supplied requirements, so BVA is not applicable to them.

OTP

Boundary	Why it is a boundary	Why it should be tested
Just below minimum: 5 digits	One less than the required 6-digit length	Verifies that the system rejects values below the required length
Minimum: 6 digits	Exact required length	Confirms the smallest valid OTP is accepted
Just above minimum: 7 digits	One more than the required length	Verifies that the system rejects values above the allowed fixed length
Nominal value: 6 digits	Standard valid OTP value	Confirms the normal successful path
Just below maximum: 5 digits	Same as the lower-bound side because the valid range is fixed at 6	Confirms the lower edge of the fixed-length rule
Maximum: 6 digits	Exact maximum allowed because the rule is fixed-length	Confirms the upper edge of the fixed-length rule

Just above maximum: 7 digits	One more than the permitted length	Confirms that values beyond the upper bound are rejected
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New password

Boundary	Why it is a boundary	Why it should be tested
Just below minimum: 7 characters	One character shorter than the minimum required length	Verifies that passwords below the minimum length are rejected
Minimum: 8 characters	Smallest accepted length by the stated rule	Confirms the minimum acceptable password length is accepted
Just above minimum: 9 characters	One character longer than the minimum	Confirms that slightly longer valid passwords are accepted
Nominal value: 8+ characters with all required classes	Typical valid strong password	Confirms normal successful behavior
Just below maximum: Not defined	No maximum length is provided	No upper-bound boundary can be derived
Maximum: Not defined	No maximum length is provided	No upper-bound boundary can be derived
Just above maximum: Not defined	No maximum length is provided	No upper-bound boundary can be derived

Email address

- No measurable numeric or length boundary is defined.
- Boundary Value Analysis is not applicable.

Confirm new password

- The requirement is equality-based rather than length-based.
- Boundary Value Analysis is not applicable.

6. Test cases for boundary value analysis

Test Case ID	Input(s)	Test Data	Boundary Tested	Expected Result	Test Type
BVA-01	OTP	12345	<6 <6	Reject OTP because it is shorter than the required 6-digit length	Lower Boundary
BVA-02	OTP	123456	=6	Accept OTP because it matches the required length	Lower Boundary

			=6		
BVA-0 3	OTP	1234567	>6 >6	Reject OTP because it exceeds the required 6-digit length	Upper Boundary
BVA-0 4	OTP	123456	Nominal	Accept OTP as a standard valid value	Nominal
BVA-0 5	OTP	12345	<6 <6	Reject OTP because it is below the fixed-length lower/upper boundary	Upper Boundary
BVA-0 6	OTP	123456	=6 =6	Accept OTP because it is the exact maximum allowed length	Upper Boundary
BVA-0 7	OTP	1234567	>6 >6	Reject OTP because it is beyond the maximum allowed length	Upper Boundary
BVA-0 8	New password	A1!bcde	<8 <8 characters	Reject password because it is shorter than the minimum required length	Lower Boundary
BVA-0 9	New password	A1!bcdef	=8 =8 characters	Accept password because it meets the minimum required length and strength rules	Lower Boundary
BVA-1 0	New password	A1!bcdefg	>8 >8 characters	Accept password because it is longer than the minimum and still satisfies strength rules	Lower Boundary
BVA-1 1	New password	StrongPass1!	Nominal	Accept password as a typical valid strong password	Nominal

II. Pool B: FR-10: Order state machine

1. Requirement Analysis

Functional requirements summary

FR-10 defines the order state machine for the EShop system.

The business rules are:

- An order has 5 valid states:
 - pending
 - confirmed

- shipping
- delivered
- canceled
- The allowed transitions follow the defined workflow:
 - pending → confirmed
 - confirmed → shipping
 - shipping → delivered
 - pending → canceled
 - confirmed → canceled
- Final states:
 - delivered
 - canceled
- Once an order reaches a final state, it cannot transition to any other state.
- If the order is in shipping, only Admin may cancel it; a regular User cannot cancel it.
- Any invalid transition must be rejected with an error message.

User inputs

The relevant user inputs are:

- Order ID
- Current order status
- Requested target status
- Actor role (User or Admin)

Assumptions

- The system stores order status as a string value.
- The actor role is derived from the authenticated user context.
- The system checks both the current status and the actor role before allowing a transition.
- The requirement does not define a separate field for “reason for change,” so it is not considered an input in this analysis.

2. Input Domain Identification

Input name	Data type	Valid domain	Invalid domain	Constraints	Dependencies	Assumptions
Order ID	String / identifier	Existing order record	Non-existing order, empty, malformed, null	Must reference a valid existing order	Depends on stored order data	The system can resolve the order by ID
Current order status	String	pending, confirmed, shipping,	Unknown/unsupported status,	Must be one of the defined states	Depends on the order record	Current status is read

		delivered, canceled	empty, null, malformed		identified by Order ID	from the database
Requested target status	String	A valid next state allowed by the state machine for the current status and actor	Invalid state, disallowed transition, empty, null	Must be one of the defined states and must be allowed by the rules	Depends on current order status and actor role	Transition validity is checked against the state machine
Actor role	String	admin, user	unauthenticated, unknown, empty, null	Must be a recognized role	Depends on authentication context	Admin vs User affects cancellation permission in shipping

3. Equivalence Class Partitioning

Order ID

EC ID	Equivalence class	Type	Why it exists
EC1	Existing order ID	Valid	Represents a normal, valid order reference
EC2	Non-existing order ID	Invalid	Ensures unknown orders are rejected
EC3	Empty / null / missing order ID	Invalid	Covers missing identifier input
EC4	Malformed / invalid format order ID	Invalid	Covers unsupported or malformed identifiers

Current order status

EC ID	Equivalence class	Type	Why it exists
EC1	Status = pending	Valid	Represents the starting state
EC2	Status = confirmed	Valid	Represents a middle state
EC3	Status = shipping	Valid	Represents a special state for cancellation rules
EC4	Status = delivered	Valid	Represents a final state
EC5	Status = canceled	Valid	Represents another final state

EC6	Status = unknown / unsupported / empty / null	Invalid	Covers malformed or unavailable status values
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Requested target status

EC ID	Equivalence class	Type	Why it exists
EC 1	Requested status is a valid allowed next state	Valid	Represents a legal transition
EC 2	Requested status is a disallowed transition for the current state	Invalid	Covers invalid state changes
EC 3	Requested status attempts to change from a final state	Invalid	Covers the final-state restriction
EC 4	Requested status is an unsupported/unknown value	Invalid	Covers invalid or misspelled status values
EC 5	Requested status is empty / null / missing	Invalid	Covers missing request input

Actor role

EC ID	Equivalence class	Type	Why it exists
EC1	Actor role = admin	Valid	Represents the privileged role
EC2	Actor role = user	Valid	Represents the standard role
EC3	Actor role = unauthenticated / unknown / empty / null	Invalid	Covers invalid or missing role context